

which are still equations of considerable generality. If now $\phi = \theta$ and U_θ is a function of $m + n + p + \dots$ of the order θ , the quantity $\sigma\{\theta^{\theta-1}\Delta(\theta)\}$ reduces itself to the single term of U_θ which contains the product $mnp\dots$. Hence, if

$$U_\theta = [a + m + n + p\dots, \theta]$$

in which afterwards $a = r - m - n - p - \dots$, we have the formula (A). Again, if $\phi = \theta + 1$, and $U_{\theta+1}$ is a function of $m + n + p\dots$ of the order θ , the sum $\sigma\{\theta^{\theta+1}\Delta(\phi)\}$ vanishes; whence writing $U_{\theta+1} = [m + n + p\dots - \theta, \theta]$, we have the formula (B). Similarly, if in the second formula $\phi = \theta + 1$, and $U_{\theta+1}$ is a function of $m + n + p\dots$ of the degree θ , then

$$\sigma\{(\theta + 1 - g)\theta^{\theta-g}\Delta(\theta + 1)\}$$

reduces itself to the term which contains $mn\dots + np\dots + mp\dots + \&c.$; whence, if

$$U_{\theta+1} = [m + n + p + \dots - \theta, \theta],$$

we have the formula (C).

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41.

ON CERTAIN FORMULÆ FOR DIFFERENTIATION WITH APPLICATIONS TO THE EVALUATION OF DEFINITE INTEGRALS.

[From the *Cambridge and Dublin Mathematical Journal*, vol. II. (1847), pp. 122–128.]

In attempting to investigate a formula in the theory of multiple definite integrals (which will be noticed in the sequel), I was led to the question of determining the $(i + 1)^{\text{th}}$ differential coefficient of the $2i^{\text{th}}$ power of $\sqrt{(x + \lambda)} - \sqrt{(x + \mu)}$; the only way that occurred for effecting this was to find the successive differential coefficients of this quantity, which may be effected as follows. Assume

$$U_{k,i} = \{(x + \lambda)(x + \mu)\}^{\frac{1}{2}k} \{\sqrt{(x + \lambda)} - \sqrt{(x + \mu)}\}^{-i};$$

then

$$\begin{aligned} \frac{1}{U_{k,i}} \frac{d}{dx} U_{k,i} &= \frac{1}{2}k \frac{2x + \lambda + \mu}{(x + \lambda)(x + \mu)} + \frac{i}{\sqrt{\{(x + \lambda)(x + \mu)\}}} \\ &= \frac{1}{2}k \frac{[\sqrt{(x + \lambda)} + \sqrt{(x + \mu)}]^2 - 2\sqrt{\{(x + \lambda)(x + \mu)\}}}{(x + \lambda)(x + \mu)} + \frac{i}{\sqrt{\{(x + \lambda)(x + \mu)\}}} \\ &= \frac{1}{2}k \frac{(\lambda - \mu)^2}{\{\sqrt{(x + \lambda)} - \sqrt{(x + \mu)}\}^2(x + \lambda)(x + \mu)} + \frac{k + i}{\sqrt{\{(x + \lambda)(x + \mu)\}}}; \end{aligned}$$

or, attending to the signification of $U_{k,i}$,

$$\frac{d}{dx} U_{k,i} = \frac{1}{2}k(\lambda - \mu)^2 U_{k-2,i-2} - (k + i) U_{k-1,i}.$$

Hence

$$\frac{1}{i} \frac{d}{dx} U_{0,i} = U_{-1,i}.$$

$$\frac{1}{i} \frac{d^2}{dx^2} U_{0,i} = \frac{1}{2}(\lambda - \mu)^2 U_{-2,i-2} + (i - 1) U_{-2,i};$$

&c.

from which the law is easily seen to be of the form

$$\frac{(-)^r}{i} \left(\frac{d}{dx} \right)^r U_{0,i} = S_\theta K_{r,\theta} (\lambda - \mu)^{2r-2-2\theta} U_{-2r+\theta+\theta, i-r+\theta}$$

(where the extreme values of θ are 0 and $(r-1)$ respectively) and $K_{r,\theta}$ is determined by

$$K_{r+1,\theta+1} = (r-1-\frac{1}{2}\theta) K_{r,\theta+1} + (i-3r+2+2\theta) K_{r,\theta}.$$

This equation is satisfied by

$$K_{r,\theta} = \frac{\Gamma(r-\frac{1}{2}-\theta) \Gamma(2r-1-\theta) \Gamma(i-r+\theta+1)}{\Gamma(\frac{1}{2}) \Gamma(\theta+2) \Gamma(2r-1-2\theta) \Gamma(i-r+1)};$$

for in the first place this gives

$$\begin{aligned} (r-1-\frac{1}{2}\theta) K_{r,\theta+1} &= \frac{(r-1-\frac{1}{2}\theta) \Gamma(r-\frac{3}{2}-\theta) \Gamma(2r-2-\theta) \Gamma(i-r+\theta+2)}{\Gamma(\frac{1}{2}) \Gamma(\theta+2) \Gamma(2r-3-2\theta) \Gamma(i-r+1)} \\ &= \frac{\Gamma(r-\frac{1}{2}-\theta) \Gamma(2r-1-\theta) \Gamma(i-r+\theta+2)}{\Gamma(\frac{1}{2}) \Gamma(\theta+2) \Gamma(2r-2-2\theta) \Gamma(i-r+1)}, \end{aligned}$$

and hence the second side of the equation reduces itself to

$$\frac{\Gamma(r-\frac{1}{2}-\theta) \Gamma(2r-1-\theta) \Gamma(i-r+\theta+1)}{\Gamma(\frac{1}{2}) \Gamma(\theta+2) \Gamma(2r-1-2\theta) \Gamma(i-r+1)} \{2(r-1-\theta)(i-r+\theta+1) + (\theta+1)(i-3r+2+2\theta)\},$$

where the quantity within brackets reduces itself to $(i-r)(2r-1-\theta)$, so that the above value reduces itself to $K_{r+1,\theta+1}$, which verifies the equation in question. Also by comparing the first few terms, it is immediately seen that the above is the correct value of $K_{r,\theta}$, so that

$$\frac{(-)^{r+1}}{i} \left(\frac{d}{dx} \right)^r U_{0,i} = S_\theta \frac{\Gamma(r-\frac{1}{2}-\theta) \Gamma(2r-1-\theta) \Gamma(i-r+\theta+1)}{\Gamma(\frac{1}{2}) \Gamma(\theta+2) \Gamma(2r-1-2\theta) \Gamma(i-r+1)} (\lambda - \mu)^{2r-2-2\theta} U_{-2r+\theta+\theta, i-r+\theta} \dots (1),$$

θ extending as before from 0 to $(r-1)$. In particular if i be integer and $r = i+1$,

$$\frac{(-)^{i+1}}{i} \left(\frac{d}{dx} \right)^{i+1} \{\sqrt{(x+\lambda)} - \sqrt{(x+\mu)}\}^{2i} = \frac{\Gamma(i+\frac{1}{2})}{\Gamma(\frac{1}{2})} (\lambda - \mu)^{2i} \frac{1}{\{(x+\lambda)(x+\mu)\}^{i+\frac{1}{2}}} \dots (2),$$

(since the factor $\Gamma(i-r+\theta+1) \div \Gamma(i-r+1)$ vanishes except for $\theta = 0$ on account of $\Gamma(i-r+1) = \infty$). Thus also, if r be greater than $(i+1)$, $= i+1+s$ suppose, then

$$(-)^r \left(\frac{d}{dx} \right)^r \frac{1}{\{(x+\lambda)(x+\mu)\}^{i+\frac{1}{2}}} = S_\theta \frac{\Gamma(i+s+\frac{1}{2}-\theta) \Gamma(2i+2s+1-\theta) \Gamma(\theta-s)}{\Gamma(\frac{1}{2}) \Gamma(\theta+1) \Gamma(2i+2s+1-2\theta) \Gamma(-s)} (\lambda - \mu)^{2s-2\theta} U_{-2i-2s+\theta+\theta, -s+\theta} \dots (3),$$

where θ extends only from $\theta = 0$ to $\theta = s$, on account of the factor $\Gamma(\theta-s) \div \Gamma(-s)$, which vanishes for greater values of θ : a rather better form is obtained by replacing this factor by

$$(-)^\theta \frac{\Gamma(1+s)}{\Gamma(1+s-\theta)}.$$

The above formulae have been deduced on the supposition of i being an integer; assuming that they hold generally, the equation (2) gives, by writing $(i-\frac{1}{2})$ for i ,

$$\frac{(-)^{i+\frac{1}{2}}}{i-\frac{1}{2}} \left(\frac{d}{dx} \right)^{i+\frac{1}{2}} \{\sqrt{(x+\lambda)} - \sqrt{(x+\mu)}\}^{2i-1} = \frac{\Gamma i}{\Gamma(\frac{1}{2})} (\lambda - \mu)^{2i-1} \frac{1}{\{(x+\lambda)(x+\mu)\}^{i+\frac{1}{2}}},$$

or integrating $(i + \frac{1}{2})$ times by means of the formula

$$\int_0^\infty x^{a-1} f x dx = \frac{\Gamma(i + \frac{1}{2})}{(-)^{i+\frac{1}{2}}} \left(\int_0^\infty dx \right)^{i+\frac{1}{2}} f x, \quad x = 0;$$

this gives

$$\int_0^\infty \frac{x^{i-1} dx}{\{(x + \lambda)(x + \mu)\}^{i+\frac{1}{2}}} = \frac{\Gamma\frac{1}{2} \Gamma(i - \frac{1}{2})}{\Gamma i} \frac{1}{(\sqrt{\lambda} + \sqrt{\mu})^{2i-1} \sqrt{\mu}} \dots (4),$$

whence also

$$\int_0^\infty \frac{x^{i-1} dx}{(x + \lambda)^{i+\frac{1}{2}} (x + \mu)^{i+\frac{1}{2}}} = \frac{\Gamma\frac{1}{2} \Gamma(i + \frac{1}{2})}{\Gamma(i + 1)} \frac{1}{(\sqrt{\lambda} + \sqrt{\mu})^{2i} \sqrt{\lambda} \sqrt{\mu}} \dots (5);$$

and from these, by simple transformations,

$$\int_\beta^\alpha \frac{(\alpha - x)^{i-1} (x - \beta)^{i-1} dx}{\{(\alpha - x) + m(x - \beta)\}^i} = \frac{\Gamma\frac{1}{2} \Gamma(i + \frac{1}{2})}{\Gamma(i + 1)} \frac{(\alpha - \beta)^i}{(\sqrt{m} + 1)^{2i}} \dots (6),$$

$$\int_\beta^\alpha \frac{(\alpha - x)^{i-1} (x - \beta)^{i-1} dx}{\{(\alpha - x) + m(x - \beta)\}^i} = \frac{\Gamma\frac{1}{2} \Gamma(i - \frac{1}{2})}{\Gamma i} \frac{(\alpha - \beta)^{i-1}}{(\sqrt{m} + 1)^{2i-1}} \dots (7).$$

These last two formulae are connected also by the following general property:

“If

$$(a, b, i) = \int_\beta^\alpha \frac{(\alpha - x)^{a-1} (x - \beta)^{b-1} dx}{[(\alpha - x) + m(x - \beta)]^i},$$

then

$$(a, b, i) = \frac{\Gamma a \Gamma b}{\Gamma(a + b - i) \Gamma i} (\alpha - \beta)^{a+b-i-1} (a + b - i, i, \beta) \dots (8),$$

which I have proved by means of a multiple¹ integral. From (6) we may obtain for $\gamma < 1$,

$$\int_{-1}^1 \frac{(1 - x^2)^{i-\frac{1}{2}} dx}{(1 - 2\gamma x + \gamma^2)^i} = \frac{\Gamma\frac{1}{2} \Gamma i}{\Gamma(i + 1)} \frac{1}{(1 - \gamma^2)^{i-\frac{1}{2}}} \dots (9),$$

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which however is only a particular case of

$$\int_{-1}^1 dx (1 - x^2)^{i-\frac{1}{2}} (1 - 2\gamma x + \gamma^2)^{-\theta} \left(\frac{d}{d\beta} \right)^i \left[\beta^i \left(1 - 2\frac{\beta}{\gamma} x + \frac{\beta^2}{\gamma^2} \right)^{-\theta} \right] = \frac{\Gamma\frac{1}{2} \Gamma(i + \frac{1}{2})}{\Gamma(i + 1)} \beta^{i-i} (1 - \beta)^{-2i} \dots (10),$$

which supposes γ and $\frac{\beta}{\gamma}$ each less than unity. This formula was obtained in the case of $(i + \frac{1}{2})$ an integer, from a theorem, *Leg. Cal. Int.*, tom. II. p. 258, but there is no doubt that it is generally true.

From (9), by writing $x = \cos \theta$, we have

$$\int_0^\pi \frac{\sin^{2i} \theta d\theta}{(1 - 2\gamma \cos \theta + \gamma^2)^i} = \frac{\Gamma\frac{1}{2} \Gamma(i + \frac{1}{2})}{\Gamma(i + 1)} \dots (11),$$

¹[The triple integral $\iiint x^{a-1} y^{b-1} z^{i-1} e^{-(\alpha-x)x - (\beta-y)y - m(x-\beta)z} dx dy dz$.]

¹This is immediately transformed into

$$\int_0^\infty \frac{x^{i-1} dx}{(ax^2 + bx + c)^i} = \frac{\Gamma\frac{1}{2} \Gamma(i - \frac{1}{2})}{\Gamma i} \frac{1}{\{b + 2\sqrt{(ac)}\}^{i-\frac{1}{2}}},$$

which is a particular case of a formula which will be demonstrated in a subsequent paper. [I am not sure to what this refers.]

which may also be demonstrated by the common equation in the theory of elliptic functions $\sin(\phi - \theta) = \gamma \sin \phi$, as was pointed out to me by Mr [Sir W.] Thomson. It may be compared with the following formula of Jacobi's, *Crelle*, tom. XV. [1836] p. 7,

$$\int_0^\pi \frac{\sin^{2i-1} \theta \, d\theta}{(1 - 2\gamma \cos \theta + \gamma^2)^i} = \frac{1}{\Gamma(i + \frac{1}{2})} \int_0^\pi \frac{\cos(i - \frac{1}{2})\theta \, d\theta}{\sqrt{(1 - 2\gamma \cos \theta + \gamma^2)}} \dots (12).$$

Consider the multiple integral

$$W = \int \frac{dx \, dy \dots}{\{(x - a)^2 + \dots + u^2\}^i} \dots (13),$$

the number of variables being $(2i + 1)$ (not necessarily odd), and the equation of the limits being

$$x^2 + y^2 \dots = \xi;$$

then, as will presently be shown, W may be expanded in the form

$$W = \pi^{i+\frac{1}{2}} S_\lambda \frac{(-)^\lambda A^\lambda}{2^{2\lambda} \Gamma(\lambda + 1) \Gamma(i + \lambda + \frac{1}{2})} \left(\frac{d}{du} \right)^{2\lambda} \int_0^\xi \xi^{i-\frac{1}{2}} (\xi + u^2)^{-i} \, d\xi \dots (14),$$

where $A = a^2 + b^2 + \dots$, and λ extends from 0 to ∞ . Suppose next

$$V = \int \frac{dx \, dy \dots}{\{(x - a)^2 \dots + v^2\}^i (x^2 + \dots + v^2)^{i+1}} \dots (15),$$

the number of variables as before, and the limits for each variable being $-\infty, \infty$. We have immediately

$$V = \int_0^\infty \frac{1}{(\xi + v^2)^{i+1}} \frac{dW}{d\xi} \, d\xi,$$

W as before, i.e.

$$V = \pi^{i+\frac{1}{2}} S_\lambda \frac{(-)^\lambda A^\lambda}{2^{2\lambda} \Gamma(\lambda + 1) \Gamma(i + \lambda + \frac{1}{2})} \left(\frac{d}{du} \right)^{2\lambda} \int_0^\infty \frac{\xi^{i-\frac{1}{2}} \, d\xi}{(\xi + v^2)^i (\xi + u^2)^{i+1}}.$$

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But writing u^2, v^2 for λ, μ in the formula (5) (u and v being supposed positive), the integral in this formula is

$$\frac{\sqrt{\pi} \Gamma(i + \frac{1}{2})}{\Gamma(i + 1)} \frac{1}{v(u + v)^{2i}};$$

hence, after a slight reduction,

$$V = \frac{\pi^{i+1}}{v \Gamma(i + 1)} S_\lambda \frac{(-)^\lambda \Gamma(i + \lambda + 1)}{\Gamma(i + 1) \Gamma(\lambda + 1)} \frac{A^\lambda}{(u + v)^{2\lambda}} \frac{1}{u} \dots$$

or finally

$$V = \frac{\pi^{i+1}}{\Gamma(i + 1) v} \frac{1}{\{(u + v)^2 + A\}^{i+1}} \dots (16),$$

a remarkable formula, the discovery of which is due to Mr Thomson. It only remains to prove the formula for W . Out of the variety of ways in which this may be accomplished, the following is a tolerably simple one. In the first place, by a linear transformation corresponding to that between two sets of rectangular axes, we have

$$W = \int \frac{dx \, dy \dots}{\{(x - \sqrt{A})^2 + y^2 \dots + u^2\}^i};$$

or expanding in powers of A , and putting for shortness $R = x^2 + y^2 \cdots + u^2$, the general term of W is

$$(-)^\sigma A^\sigma \frac{\Gamma(i + \lambda + \sigma)}{\Gamma\sigma \Gamma(\lambda - \sigma + 1) \Gamma(2\sigma + 1)} 2^{2\sigma} \int x^{2\sigma} R^{-i-\lambda-\sigma} dx dy \dots$$

the limits being as before $x^2 + y^2 + \cdots = \xi$. To effect the integrations, write $\sqrt{\xi}\sqrt{x}$, $\sqrt{\xi}\sqrt{y}$, &c. for $x, y \dots$ so that the equation of the limits becomes $x + y + \cdots = 1$. Also restricting the integral to positive values, we must multiply it by 2^{2i+1} ; the integral thus becomes

$$\xi^{\sigma+i+\frac{1}{2}} \int x^{\sigma-\frac{1}{2}} y^{-\frac{1}{2}} \dots \{\xi(x + y \dots) + u^2\}^{-i-\lambda-\sigma} dx dy \dots$$

equivalent to

$$\xi^{\sigma+i+\frac{1}{2}} \frac{\Gamma(\sigma + \frac{1}{2})\pi^i}{\Gamma(i + \sigma + \frac{1}{2})} \int_0^1 \theta^{i+\sigma-\frac{1}{2}} (\xi\theta + u^2)^{-i-\lambda-\sigma} d\theta;$$

i.e. to

$$\frac{\Gamma(\sigma + \frac{1}{2})\pi^i}{\Gamma(i + \sigma + \frac{1}{2})} \int_0^\xi \xi^{i+\sigma-\frac{1}{2}} (\xi + u^2)^{-i-\lambda-\sigma} d\xi.$$

Hence, after a slight reduction, the general term of W is

$$\frac{\pi^{i+\frac{1}{2}}}{\Gamma i} (-)^\sigma A^\sigma \frac{\Gamma(i + \lambda + \sigma)}{\Gamma(\sigma + 1) \Gamma(\lambda - \sigma + 1) \Gamma(i + \sigma + \frac{1}{2})} \int_0^\xi \xi^{i+\sigma-\frac{1}{2}} (\xi + u^2)^{-i-\lambda-\sigma} d\xi,$$

where σ may be considered as extending from 0 to λ inclusively, and then λ from 0 to ∞ . But by a formula easily proved

$$\left(\frac{d}{du}\right)^{2\lambda} (\xi + u^2)^{-i} = \frac{2^{2\lambda} \Gamma(\lambda + 1) \Gamma(i + \lambda + \frac{1}{2})}{\Gamma i} \times S (-)^\sigma \frac{\Gamma(i + \lambda + \sigma)}{\Gamma(\sigma + 1) \Gamma(\lambda - \sigma + 1) \Gamma(i + \sigma + \frac{1}{2})} \xi^\sigma (\xi + u^2)^{-i-\lambda-\sigma},$$

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where σ extends from 0 to λ . Hence, substituting and prefixing the summatory sign,

$$W = \pi^{i+\frac{1}{2}} S \frac{(-)^\lambda A^\lambda}{2^{2\lambda} \Gamma(\lambda + 1) \Gamma(i + \lambda + \frac{1}{2})} \left(\frac{d}{du}\right)^{2\lambda} \int_0^\xi \xi^{i-\frac{1}{2}} (\xi + u^2)^{-i} d\xi,$$

where λ extends from 0 to ∞ , the formula required.

[I annex the following Note added in MS. in my copy of the *Journal*, and referring to the formula, ante p. 267; a is written to denote $\lambda - \mu$.

N.B. It would be worth while to find the general differential coefficient of $U_{k,i}$.

$$\partial_x U_{k,i} = -(k + i) U_{k-1,i} + \frac{1}{2} k a^2 U_{k-2,i-2}$$

from which it is easy to see that

$$\begin{aligned} \partial_x^r U_{k,i} &= (-)^r [K_{r,0} U_{k-r,i} \\ &\quad + (-)^{r-1} K_{r,1} a^2 U_{k-r-2,i-2} \\ &\quad + \dots \\ &\quad + K_{r,r} a^{2r} U_{k-2r,i-r}]. \end{aligned}$$

The general term of $\partial_x^{r+1} U_{k,i}$ is

$$\begin{aligned} & (-)^{r-\theta} K_{r,\theta} a^{2\theta} \left[\frac{1}{2} (k-r-\theta) a^2 U_{k-r-\theta-2, i-\theta-2} \right] \\ & + (-)^{r-\theta-1} K_{r,\theta+1} a^{2\theta+2} \left[-(k+i-2\theta-2-r) U_{k-r-\theta-2, i-\theta-1} \right], \end{aligned}$$

which must be equal to

$$(-)^{r-\theta} K_{r+1,\theta+1} a^{2\theta+2} U_{k-r-\theta-2, i-\theta-1};$$

therefore

$$K_{r+1,\theta+1} = (k+i-r-2\theta-2) K_{r,\theta+1} + \frac{1}{2} (k-r-\theta) K_{r,\theta}.$$

In particular

$$\begin{aligned} K_{r+1,0} &= -(k+i-r) K_{r,0} = 0, \\ K_{r+1,1} &= -(k+i-r-2) K_{r,1} + \frac{1}{2} (k-r) K_{r,0}, \\ &\vdots \\ K_{r+1,r+1} &= \frac{1}{2} (k-r) K_{r,r}, \\ K_{r,0} &= -[k+i]^r, \\ K_{r,1} &= -[r \{k^2 + (i-r)k - \frac{1}{2}(r-1)i\} [k+i-2]^{r-2}], \\ K_{r,r} &= -[\frac{1}{2}k]^r, \end{aligned}$$

whence

$$K_{r,1} = kr \{k^2 + (i-r)k - \frac{1}{2}(r-1)i\} [k+i-2]^{r-2},$$

which appears to indicate a complicated general law.

Even the verification of $K_{r,1}$ is long, thus the equation becomes

$$\frac{r+1}{r} [k+i-2]^{r-1} \{k^2 + (i-r-1)k - \frac{1}{2}ri\} - (k+i-r-2)r \{k^2 + (i-r)k - \frac{1}{2}(r-1)i\} [k+i-2]^{r-2} = (i-r) [k+i]^r,$$

or

$$\frac{r+1}{r} (k+i-r) \{k^2 + (i-r-1)k - \frac{1}{2}ri\} - r(k+i-r-2) \{k^2 + (i-r)k - \frac{1}{2}(r-1)i\} = (i-r)(k+i)(k+i-1),$$

which is identical, as may be most easily seen by taking first the coefficient of k^3 , and then writing $k=r$, $k=-i$, $k=-i-1$.]

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42.

ON THE CAUSTIC BY REFLECTION AT A CIRCLE.

[From the *Cambridge and Dublin Mathematical Journal*, vol. II. (1847), pp. 128–130.]

The following solution of the problem is that given by M. de St-Laurent (*Annales de Gergonne*, t. XVII. [1826] pp. 128–134); the process of elimination is somewhat different.

The centre of the circle being taken for the origin, let k be its radius; a, b the coordinates of the luminous point; ξ, η those of the point at which the reflection takes place; x, y those of any point in the reflected ray: we have in the first place

$$\xi^2 + \eta^2 = k^2 \quad \dots \quad (1).$$

There is no difficulty in finding the equation of the reflected ray²; this is

$$(b\xi - a\eta)(\xi x + \eta y - k^2) + (y\xi - x\eta)(a\xi + b\eta - k^2) = 0 \quad \dots \quad (2),$$

²To do this in the simplest way, write

$$\rho^2 = (\xi - x)^2 + (\eta - y)^2, \quad \sigma^2 = (\xi - a)^2 + (\eta - b)^2,$$

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or, arranging the terms in a more convenient order,

$$(bx + ay)(\xi^2 - \eta^2) + 2(by - ax)\xi\eta - k^2(b + y)\xi + k^2(a + x)\eta = 0 \quad \dots \quad (2').$$

Hence, considering ξ, η as indeterminate parameters connected by the equation (1), the locus of the curve generated by the continued intersections of the lines (2) will be found by eliminating ξ, η, λ from these equations and the system

$$\xi [\lambda + 2(bx + ay)] + \eta [2(by - ax)] - k^2(b + y) = 0 \quad \dots \quad (3),$$

$$\xi [2(by - ax)] + \eta [\lambda - 2(bx + ay)] + k^2(a + x) = 0 \quad \dots \quad (4),$$

and from these, multiplying by ξ, η , adding and reducing by (2), we have

$$-\xi(b + y) + \eta(a + x) - \lambda = 0 \quad \dots \quad (5),$$

which replaces the equation (2) or (2'). Thus the equations from which ξ, η, λ are to be eliminated are (1), (3), (4), (5).

From (3), (4), (5), by the elimination of ξ, η , we have

$$\begin{aligned} & -\lambda [\lambda^2 - 4(bx + ay)^2] - 4\lambda(by - ax)(a + x)(b + y) \\ & - k^2(a + x)^2 [\lambda + 2(bx + ay)] \\ & - k^2(b + y)^2 [\lambda - 2(bx + ay)] \\ & + 4\lambda(by - ax)^2 = 0 \quad \dots \quad (6), \end{aligned}$$

or, reducing,

$$-\lambda^3 + \lambda \{4(a^2 + b^2)(x^2 + y^2) - k^2[(a + x)^2 + (b + y)^2]\} - 2k^2(bx - ay)(x^2 + y^2 - a^2 - b^2) = 0 \quad \dots \quad (7);$$

which may be represented by

$$-\lambda^3 + \lambda Q - 2R = 0 \quad \dots \quad (7').$$

Again, from the equations (4), (3), transposing the last terms and adding the squares, also reducing by (1),

$$k^2[(a + x)^2 + (b + y)^2] = k^2\lambda^2 + 4k^2(a^2 + b^2)(x^2 + y^2) + 4\lambda[(\xi^2 - \eta^2)(bx + ay) + 2\xi\eta(by - ax)] \quad \dots \quad (8);$$

then, by the condition of reflection,

$$\rho + \sigma = \min.,$$

ρ, σ being considered as functions of the variables ξ, η , which are connected by the equation (1). Hence

$$\frac{\xi - x}{\rho} + \frac{\xi - a}{\sigma} + \lambda\xi = 0, \quad \frac{\eta - y}{\rho} + \frac{\eta - b}{\sigma} + \lambda\eta = 0;$$

or, eliminating λ ,

$$\eta \frac{\xi - x}{\rho} - \xi \frac{\eta - y}{\rho} + \eta \frac{\xi - a}{\sigma} - \xi \frac{\eta - b}{\sigma} = 0,$$

whence

$$(\eta x - \xi y)^2 [(\xi - a)^2 + (\eta - b)^2] = (\eta a - \xi b)^2 [(\xi - x)^2 + (\eta - y)^2].$$

This may be written

$$\begin{aligned} & \{(\eta x - \xi y)(\xi - a) - (\eta a - \xi b)(\xi - x)\} [(\eta x - \xi y)(\xi - a) + (\eta a - \xi b)(\xi - x)] \\ & + \{(\eta x - \xi y)(\eta - b) - (\eta a - \xi b)(\eta - y)\} [(\eta x - \xi y)(\eta - b) + (\eta a - \xi b)(\eta - y)] = 0; \end{aligned}$$

the factors in $\{ \}$ reduce themselves respectively to ξP and ηP , where $P = \xi(y - \eta) - \eta(a - x) + a\eta - bx$; omitting the factor P , (which equated to zero, is the equation of the line through (a, b) and (ξ, η)), and replacing $\xi(\xi - a) + \eta(\eta - b)$ and $\xi(\xi - x) + \eta(\eta - y)$ by $k^2 - a\xi - b\eta$ and $k^2 - \xi x - \eta y$, respectively, we have the equation given above.

but from the same equations, multiplying by ξ, η and adding, also reducing by (1),

$$k^2\lambda + 2(bx + ay)(\xi^2 - \eta^2) + 4\xi\eta(by - ax) + k^2[-\xi(b + y) + \eta(a + x)] = 0 \quad \dots \quad (9),$$

or reducing by (5) and dividing by two,

$$k^2\lambda + (bx + ay)(\xi^2 - \eta^2) + 2\xi\eta(by - ax) = 0 \quad \dots \quad (10).$$

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Using this to reduce (8),

$$k^2[(a + x)^2 + (b + y)^2] = 4(a^2 + b^2)(x^2 + y^2) + 3\lambda^2 \quad \dots \quad (11),$$

or, from the value of Q ,

$$-3\lambda^2 + Q = 0 \quad \dots \quad (12),$$

which singularly enough is the derived equation of (7') with respect to λ : so that the equation of the curve is obtained by expressing that two of the roots of the equation (7') are equal. Multiplying (12) by λ and reducing by (7'),

$$-\lambda Q + 3R = 0;$$

or, combining this with (12),

$$27R^2 - Q^3 = 0;$$

whence, replacing R, Q by their values, we find

$$27k^4(bx - ay)^2(x^2 + y^2 - a^2 - b^2)^2 - \{4(a^2 + b^2)(x^2 + y^2) - k^2[(a + x)^2 + (b + y)^2]\}^3 = 0,$$

the equation of M. de St-Laurent.

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[43]

43.

ON THE DIFFERENTIAL EQUATIONS WHICH OCCUR IN DYNAMICAL PROBLEMS.

[From the *Cambridge and Dublin Mathematical Journal*, vol. II. (1847), pp. 210–219.]

Jacobi, in a very elaborate memoir, “Theoria novi multiplicatoris systemati aequationum differentialium vulgarium applicandi”³, has demonstrated a remarkable property of an extensive class of differential equations, namely, that when all the integrals of the system except a single one are known, the remaining integral can always be determined by a quadrature. Included in the class in question are, as Jacobi proceeds to show, the differential equations corresponding to any dynamical problem in which neither the forces nor the equations of condition involve the velocities; i.e. in all ordinary dynamical problems, when all the integrals but one are known, the remaining integral can be determined by quadratures. In the case where the forces and equations of condition are likewise independent of the time, it is immediately seen that the system may be transformed into a system in which the number of equations is less by unity than in the original one, and which does not involve the time, which may afterwards be determined by

³*Crelle*, t. XXVII. [1844], pp. 199–268 and t. XXIX. [1845], pp. 213–279 and 333–376. Compare also the memoir in *Liouville*, t. X. [1845], pp. 337–346.

a quadrature⁴; and, Jacobi's theorem applying to this new system, he arrives at the proposition "In any dynamical problem where the forces and equations of condition contain only the coordinates of the different points of the system, when all the integrals but two are determined, the remaining integrals may be found by quadratures only." In the following paper, which contains the demonstrations of these propositions, the analysis employed by Jacobi has been considerably varied in the details, but the leading features of it are preserved.

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§ 1. Let the variables $x, y, z, \dots$ &c. be connected with the variables $u, v, w, \dots$ by the same number of equations, so that the variables of each set may be considered as functions of those of the other set. And assume

$$dx \, dy \cdots = \nabla \, du \, dv \cdots;$$

if from the functions which equated to zero express the relations between the two sets of variables we form two determinants, the former with the differential coefficients of these functions with respect to $u, v, \dots$ and the latter with the differential coefficients of the same functions with respect to $x, y, \dots$ the quotient with its sign changed obtained by dividing the first of these determinants by the second is, as is well known, the value of the function ∇ .

Putting for shortness

$$\frac{dx}{du} = \alpha, \quad \frac{dy}{du} = \beta, \dots; \quad \frac{dx}{dv} = \alpha', \quad \frac{dy}{dv} = \beta', \dots \quad \&c.$$

and

$$\frac{du}{dx} = A, \quad \frac{dv}{dx} = B, \dots; \quad \frac{du}{dy} = A', \quad \frac{dv}{dy} = B', \dots$$

∇ is the reciprocal of the determinant formed with $A, B, \dots; A', B', \dots$, &c.; or it is the determinant formed with $\alpha, \beta, \dots; \alpha', \beta', \dots$, &c.

From the first of these forms, i.e. considering ∇ as a function of $A, B, \dots$

$$\frac{d\nabla}{dA} = -\nabla\alpha, \quad \frac{d\nabla}{dB} = -\nabla\beta, \dots; \quad \frac{d\nabla}{dA'} = -\nabla\alpha', \quad \frac{d\nabla}{dB'} = -\nabla\beta', \dots$$

where the quantities $\alpha, \beta, \dots \alpha', \beta', \dots$ and $A, B, \dots A', B', \dots$ may be interchanged provided $-\nabla$ be substituted for ∇ . (Demonstrations of these formulae or of some equivalent to them will be found in Jacobi's memoir "De determinantibus functionalibus," *Crelle*, t. XXII. [(1841) pp. 319–359].)

Hence

$$\frac{1}{\nabla} d\nabla + \alpha dA + \beta dB + \cdots + \alpha' dA' + \beta' dB' + \cdots = 0,$$

or reducing by

$$\frac{dA}{dy} = \frac{dB}{dx}, \dots; \quad \frac{dA'}{dy} = \frac{dB'}{dx}, \dots,$$

this becomes

$$\frac{1}{\nabla} d\nabla + \alpha \left(\frac{dA}{dx} dx + \frac{dB}{dy} dy + \cdots \right) + \beta \left(\frac{dA}{dy} dx + \frac{dB}{dy} dy + \cdots \right) + \cdots + \alpha' \left(\frac{dA'}{dx} dx + \frac{dB'}{dy} dy + \cdots \right) + \beta' \left(\frac{dA'}{dy} dx + \frac{dB'}{dy} dy + \cdots \right) + \cdots = 0;$$

⁴For, representing the velocities by $x', y', \dots$, the dynamical system takes the form

$$dt : dx : dy \cdots : dx' : dy' \cdots = 1 : x' : y' \cdots : X : Y \cdots,$$

and the system in question is simply $dx : dy \cdots : dx' : dy' \cdots = x' : y' \cdots : X : Y \cdots$.

or, reducing,

$$\frac{1}{\nabla} d\nabla + \left(\frac{dA}{du} + \frac{dA'}{dv} + \dots \right) dx + \left(\frac{dB}{du} + \frac{dB'}{dv} + \dots \right) dy + \dots = 0;$$

whence separating the differentials and replacing $A, A', \dots; B, B', \dots;$ by their values

$$\begin{aligned} \frac{1}{\nabla} \frac{d\nabla}{dx} + \frac{d}{du} \frac{du}{dx} + \frac{d}{dv} \frac{dv}{dx} + \dots &= 0, \\ \frac{1}{\nabla} \frac{d\nabla}{dy} + \frac{d}{du} \frac{du}{dy} + \frac{d}{dv} \frac{dv}{dy} + \dots &= 0, \end{aligned}$$

(in which $-\nabla, u, v \dots; x, y \dots$ may be substituted for $\nabla, x, y \dots; u, v \dots$).

§ 2. Let $X, Y \dots$ be any functions of the variables $x, y, \dots$ and assume

$$\begin{aligned} U &= X \frac{du}{dx} + Y \frac{dv}{dy} + \dots, \\ V &= X \frac{dv}{dx} + Y \frac{dv}{dy} + \dots, \end{aligned}$$

$U, V, \dots$ being expressed in terms of $u, v, \dots$. Then

$$\begin{aligned} \frac{dU}{du} + \frac{dV}{dv} + \dots &= X \left(\frac{d}{du} \frac{du}{dx} + \frac{d}{dv} \frac{dv}{dx} + \dots \right) + Y \left(\frac{d}{du} \frac{du}{dy} + \frac{d}{dv} \frac{dv}{dy} + \dots \right) + \dots \\ &\quad + \left(\frac{dX}{du} \frac{du}{dx} + \frac{dX}{dv} \frac{dv}{dx} + \dots \right) + \left(\frac{dY}{du} \frac{du}{dy} + \frac{dY}{dv} \frac{dv}{dy} + \dots \right) \end{aligned}$$

i.e.

$$\nabla \left(\frac{dU}{du} + \frac{dV}{dv} + \dots \right) = - \left(X \frac{d\nabla}{dx} + Y \frac{d\nabla}{dy} + \dots \right) + \nabla \left(\frac{dX}{dx} + \frac{dY}{dy} + \dots \right).$$

Also, whatever be the value of M ,

$$U \frac{dM}{du} + V \frac{dM}{dv} + \dots = X \frac{dM}{dx} + Y \frac{dM}{dy} + \dots,$$

and from these two properties,

$$\frac{dM\nabla U}{du} + \frac{dM\nabla V}{dv} + \dots = \nabla \left(\frac{dMX}{dx} + \frac{dMY}{dy} + \dots \right).$$

§ 3. Consider the system of differential equations

$$dx : dy : dz \dots = X : Y : Z \dots$$

(where, for greater clearness, an additional letter z has been introduced). From these we deduce the equivalent system

$$du : dv : dw \dots = U : V : W \dots$$